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Integrating ICT in Local Governance: A Village-Level Approach to Achieving SDGs through VPDP in Tamil Nadu

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***Abstract*— This study investigates the potential of integrating Information and Communication Technology (ICT) into the Village Panchayat Development Plan (VPDP) to strengthen decentralized rural governance and accelerate the achievement of Sustainable Development Goals (SDGs) in Tamil Nadu. Conducted in Mavuthampathy and Palathurai Panchayats in Coimbatore District, the research utilized a mixed-methods approach comprising a household survey of 400 respondents, focus group discussions, participatory rural appraisals, and key informant interviews. The statistical analysis using Jamovi, highlight several socio-economic and infrastructural challenges, particularly in livelihood, sanitation, education, and access to services. While the VPDP process enables participatory planning, its implementation is limited by low digital access and poor community awareness. This article recommends context-specific ICT solutions such as digital literacy campaigns, GIS mapping for planning, and mobile-based grievance redressal. The study demonstrates how technology may effectively support inclusive and sustainable rural development when paired with local expertise and community involvement.**

Keywords*— *VPDP, ICT4D, e-governance, rural development, participatory planning, SDGs, Tamil Nadu

I. Introduction

India's decentralized system of governance, rooted in the 73rd Constitutional Amendment Act, offers a strong platform for localizing development through Gram Panchayats. The Ministry of Panchayati Raj (MoPR) introduced the Village Panchayat Development Plan (VPDP), a participatory planning tool, to help local bodies rank development interventions, identify community needs, and implement plans that align with the Sustainable Development Goals (SDGs). [1], [2]. Due to limitations such low public awareness, poor digital infrastructure, restricted fiscal autonomy, and inadequate technical capacity at the panchayat level, VPDP implementation in many rural regions is still unsatisfactory despite its potential [3], [4]. Because of this, the planning process frequently devolves into a bureaucratic formality that does not represent the goals of the community or produce the desired results.

Information and communication technology (ICT) integration in rural governance presents a workable answer to these problems, especially in light of India's Digital India Mission and growing emphasis on e-government. Transparency, participatory monitoring, and grassroots service delivery can all be enhanced by ICT tools such as Geographic Information Systems (GIS), mobile platforms, and real-time data dashboards [5, 6].

The findings of a mixed-method study conducted in Tamil Nadu's Coimbatore District's Mavuthampathy and Palathurai Panchayats are presented in this paper. The study explores the real-world implementation of VPDPs, identifies critical governance and service gaps based on data from the community level, and recommends context-appropriate ICT interventions to improve planning and guarantee SDG achievement.

This study contributes to the body of research on ICT for rural governance by arguing for a socially grounded, village-level approach to digital integration that prioritises development equality, evidence-based decision-making, and community engagement. ICT can effectively support inclusive and sustainable rural change in local contexts [7].

II. Review of Literature

In order to connect local needs with national and international development priorities—particularly the Sustainable Development Goals (SDGs)—the Village

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Panchayat Development Plan (VPDP) was conceived as a participatory planning tool [2]. The VPDP will link national and global development agendas, including the Sustainable Development Goals (SDGs), with grassroots planning [8]. Particularly in rural India, scholarly literature emphasises that decentralised planning promotes inclusive and need-based development [1], [3]. The localisation of SDGs at the local government level is one of the main focusses of VPDP [9]. One of the main drivers of SDG localisation and rural governance is the use of technology in VPDP [10]. Despite the emphasis on participatory rural evaluation approaches, research has shown that their implementation is unequal, leading to planning and implementation gaps [11]. However, due to a lack of data-driven initiatives, low institutional capacity, and low community engagement, its implementation is still inconsistent [4].

According to research by Chatterjee [12] and the Ministry of Panchayati Raj [1], VPDPs are only successful if they are supported by tools that enable transparency, performance tracking, and real-time monitoring. This is especially important when dealing with marginalised groups that are typically left out of institutional decision-making processes, such as Scheduled Tribes (STs) and Scheduled Castes (SCs) [13]. Poor livelihood opportunities also hinder economic sustainability in rural areas, forcing individuals to migrate to metropolitan areas [14].

The use of information and communication technology (ICT) has grown in importance as a means of bridging these gaps in implementation and planning. Promising approaches to improving rural governance include Geographic Information Systems (GIS), mobile-based participatory apps, and Management Information Systems (MIS) [5, 6]. Adoption of ICT tools, according to Kumar and Menon [10], can improve resource allocation [15], reduce leaks, and enable improved monitoring of rural initiatives.

However, researchers like Sridhar and Bhat [16] warn against a technocentric mindset, especially in rural areas with limited literacy and connectivity. They support hybrid models that blend scalable digital solutions with community-based procedures. By evaluating smart village initiatives, Jain and Thakur [17] demonstrate that context-sensitive deployment, local ownership, and training are essential for success in the digital sphere.

Ramanathan [10] highlights that ICT-enabled planning should support community discourse and conventional knowledge systems, not replace them, from a social work standpoint. This is especially true in tribal communities where development strategies heavily rely on customary government.

The body of research supporting the use of ICT in rural government is strong overall. It also warns that technology should not be viewed as a top-down imposition but rather as a tool that facilitates participatory planning. By offering a solid, fact-based case for how ICT technologies might enhance the responsiveness and inclusivity of VPDPs in Tamil Nadu, this study contributes to the growing body of literature in this area.

III. Methodology

The present study uses a mixed-method study design to assess the potential and current state of Village Panchayat Development Plans (VPDPs) in advancing the Sustainable Development Goals (SDGs) at the local level. This is done by combining qualitative participatory instruments with quantitative household surveys. Two panchayats in the Madukkarai block of the Coimbatore District, Tamil Nadu—Mavuthampathy and Palathurai—were the sites of the research.

A. Theoretical Framing and Research Design

The study is founded on Amartya Sen's Capability Approach and the Sustainable Livelihoods Framework, which emphasise community involvement as development accelerators, the expansion of liberties, and the accessibility of essential services [18], [19]. Additionally, the study adheres to the participatory rural planning paradigm promoted by the People's Plan Campaign of the Ministry of Panchayati Raj [1].

B. Sampling Method and Sample Size

A systematic household survey with 400 respondents served as the study's quantitative component. Krejcie and Morgan's Table, which suggests a representative sample of 360 in a population of 5,570, was used to create the sample [20].. The sample was increased to 400 to improve robustness. A Simple Random Sampling technique was used to ensure unbiased representation across different social and economic categories.

C. Data Collection Tools

The study employed both primary and secondary data collection methods:

- Structured household surveys, using the UBA Baseline Household Survey format, captured socio-economic, sanitation, housing, water, energy, education, and employment indicators.
- Focus Group Discussions (FGDs) were held with Self-Help Group (SHG) members, youth, and farmers to capture perceptions of governance, access to public services, and suggestions for improvement.
- Participatory Rural Appraisal (PRA) techniques, such as community mapping and transect walks, were used to understand settlement-level disparities [19], [21].
- Key Informant Interviews (KIIs) were conducted with Panchayat Presidents, Secretaries, and frontline officials.
- Secondary data from existing VPDP documents, Gram Sabha records, and government dashboards were reviewed for triangulation.

All field instruments were pilot-tested, and ethical clearance was obtained from the research institution. Informed consent was taken from all respondents.

D. Data Analysis

Quantitative data were analyzed using Jamovi such as descriptive statistics (frequencies, percentages, and cross-tabulations) and interpreted in relation to relevant SDG

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indicators. Qualitative data were analyzed using thematic analysis, identifying patterns across responses related to service access, governance barriers, and digital literacy.

Themes were mapped to key SDGs (such as SDG 3 – Health, SDG 4 – Education, SDG 6 – Water and Sanitation, and SDG 16 – Peace and Institutions) to assess alignment between community needs and existing VPDP themes [15], [22].

IV. Findings and Discussion

The study revealed a range of development gaps across the two selected Panchayats—Mavuthampathy and Palathurai—despite both having functional VPDPs. These gaps are structured across major service themes such as livelihoods, housing, sanitation, education, health, digital access, and community engagement. The data highlights both quantitative deficiencies and qualitative insights into local governance and community participation.

The statistical analyses reveal significant correlations between quality of life (QOL) and a number of demographic and socio-economic variables among respondents in Mavuthampathy and Palathurai panchayats. Independent samples t-test demonstrated a significantly higher quality of life in Palathurai (M = 91.2) than in Mavuthampathy (M = 86.0), t(394) = -3.67, p <.001, possibly due to the socio-economic disadvantage of the tribal community in Mavuthampathy. Independent Samples T-test between Quality of life and Panchayat (Table 1)

Table 1: **Independent Samples T-Test**

		Statistic	df	P
Quality of Life	Student's t	-3.67	394	< .001

Note. H_a μ Mavuthampathy ≠ μ Palathurai

Group Descriptives

	Group	N	Mean	Median	SD	SE
Quality of Life	Mavuthampathy	194	86.0	88.5	13.2	0.948
	Palathurai	202	91.2	92.0	14.6	1.03

Correlation matrix showed (Table 2) high positive correlations between all QOL areas—physical health, psychological health, social relationships, and environment—highlighting their interdependence.

Table 2: Correlation Matrix between the domains of Quality of life

		Domain 1 Physical Health	Domain 2 Psychological Health	Domain 3 Social Relationships	Domain 4 Environment
Domain 1 Physical Health	Pearson's r	—			
	df	—			
	p-value	—			
	N	—			
Domain 2 Psychological Health	Pearson's r	0.826	—		
	df	396	—		
	p-value	< .001	—		
	N	398	—		
Domain 3 Social Relationships	Pearson's r	0.572	0.616	—	
	df	396	396	—	
	p-value	< .001	< .001	—	
	N	398	398	—	
Domain 4 Environment	Pearson's r	0.763	0.797	0.533	—
	df	396	396	396	—
	p-value	< .001	< .001	< .001	—
	N	398	398	398	—

A. Livelihood and Economic Security

In both panchayats, over 70% of households reported unstable and seasonal employment. Daily wage labor and subsistence farming were the most common sources of income. While Mavuthampathy benefited from better MGNREGS coverage, Palathurai suffered from delays and lack of transparency. These findings emphasize the need for ICT-based wage tracking and digital job-matching platforms to support SDG 1 (No Poverty) and SDG 8 (Decent Work and Economic Growth) [23].

B. Water, Sanitation, and Housing

Access to piped water was inconsistent. Palathurai residents reported water supply only once every two weeks, with poor water quality. In contrast, Mavuthampathy had better coverage, but some tribal hamlets still depended on tanker water. Sanitation infrastructure, especially public toilets, was absent in both panchayats, with open defecation still practiced in some areas. These trends hinder progress toward SDG 6 (Clean Water and Sanitation). Community members proposed real-time water monitoring and mobile maintenance alerts, reflecting growing demand for simple ICT solutions [24], [25].

C. Healthcare and Health Awareness

Both panchayats lacked primary health centers (PHCs) within immediate reach. Mobile health units were irregular, and awareness of entitlements under rural health schemes remained low. The need for telemedicine and SMS-based awareness campaigns was raised in FGDs. These are essential for improving rural health under SDG 3 (Good Health and Well-being) [26].

D. Education and Digital Inclusion

School enrollment was above 90% in both panchayats; however, dropout rates—especially among tribal children—were high. Reasons included economic hardship, transportation challenges, and a lack of digital learning tools. Palathurai residents reported frequent vandalism in public schools, further discouraging attendance. Both communities highlighted the need for e-learning labs, mobile libraries, and ICT-enabled teacher training, supporting SDG 4 (Quality Education) and SDG 9 (Industry, Innovation, and Infrastructure) [17].

E. Community Participation and Governance Gaps

While VPDPs were developed in both panchayats, participation was limited to a small group of influential stakeholders. Gram Sabha attendance was below 30% in Palathurai. Over 75% of households in Mavuthampathy were

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unaware of welfare schemes. The lack of grievance redressal systems, digital feedback loops, and transparent budgeting tools underscores the need for ICT-supported participatory governance. These gaps relate to SDG 16 (Peace, Justice, and Strong Institutions) [13].

F. ICT as an Enabler of SDG Localization

Respondents showed openness to digital solutions, especially mobile-based platforms for services like water billing, grievance tracking, and access to scheme information. However, infrastructural gaps (poor connectivity, digital illiteracy) must be addressed first. A simplified technology-assisted VPDP model, combining digital tools with community facilitation, emerged as a preferred alternative to top-down planning. This confirms that ICT must complement—not replace—participatory processes [15], [16].

G. Digital Financial Inclusion and Economic Empowerment

A major finding from the present study is the economic vulnerability of rural households, particularly the dependence on seasonal labor, low-value agriculture, and government wage schemes. While VPDPs account for income and employment indicators, the integration of digital financial technologies can significantly enhance rural economic resilience and transparency.

Digital Financial Inclusion (DFI) refers to the use of digital platforms to expand access to financial services—especially for populations traditionally excluded from formal finance. In rural Tamil Nadu, digital financial tools such as Aadhaar-enabled Payment Systems (AePS), Unified Payments Interface (UPI) apps, Direct Benefit Transfers (DBT), and SHG microcredit apps are increasingly relevant.

Studies show that linking Self-Help Groups (SHGs) with mobile banking platforms increases access to microloans, improves repayment tracking, and strengthens community accountability [27]. The integration of DFI into VPDPs would enable:

- Better targeting of income-generating programs;
- Transparent wage disbursal under MGNREGS and state schemes;
- Reduced dependency on informal moneylenders;
- Promotion of savings, especially among rural women.

Moreover, mobile-based platforms like Jan Dhan-Aadhaar-Mobile (JAM) trinity can empower women and elderly beneficiaries through doorstep services [28]. However, the potential of DFI remains underutilized due to digital literacy gaps, intermittent internet access, and lack of local facilitation. This calls for dedicated VPDP components that map digital finance needs, assess digital readiness, and propose village-level interventions such as financial literacy camps, digital kiosks, and SHG-led banking correspondents.

Integrating DFI into VPDPs not only strengthens economic components of SDG 1 (No Poverty) and SDG 8 (Decent Work and Economic Growth) but also enables real-time monitoring of beneficiary coverage and fund utilization, enhancing both efficiency and equity in development outcomes [29], [30].

V. Recommendations

1. Establish Panchayat-level digital resource centers to support digital literacy and local data management.
2. Integrate GIS and mobile platforms into VPDP preparation and real-time service tracking.
3. Train elected representatives and frontline workers on digital governance tools, including MIS dashboards and scheme information systems.
4. Institutionalize citizen feedback mechanisms via low-bandwidth apps or SMS platforms.
5. Promote policy-level integration of technology-supported VPDP frameworks at the state and district levels.

This paper contributes a scalable, community-rooted framework that blends participatory governance with technology trends, offering a strategic path toward inclusive rural development and SDG realization.

VI. Conclusion

This study examined the role of the Village Panchayat Development Plan (VPDP) in promoting participatory governance and Sustainable Development Goal (SDG) localization in two rural Panchayats of Tamil Nadu. Through household surveys, participatory rural appraisal techniques, and stakeholder interviews, it revealed significant disparities in service delivery, community awareness, and the implementation quality of VPDPs.

Although VPDPs provide an institutional platform for community-based planning, their effectiveness is hindered by limited citizen participation, digital exclusion, and weak monitoring mechanisms. The research findings underscore the importance of integrating context-specific Information and Communication Technology (ICT) tools—such as GIS mapping, mobile grievance redressal systems, telemedicine services, and e-learning platforms—into the local planning process.

However, for ICT to be meaningful at the grassroots level, it must be accessible, inclusive, and complemented by capacity-building measures. Technology should not replace participatory processes but should enhance transparency, responsiveness, and efficiency in planning and service delivery. Importantly, community engagement, local ownership, and decentralized decision-making must remain central to any tech-driven intervention in rural governance.

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